

Bitcoin Exit Dominance in Monetary Coordination Games

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Abstract

Abstract. In a multipolar world with no trusted monetary coordinator, how do rational actors settle large-value transactions across trust boundaries? This paper models the choice as a non-cooperative game (the “Exit Game”) in which capital allocators choose between capturable settlement systems (“Stay”) and neutral settlement (“Exit”). The model rests on four assumptions: persistent multipolarity, rational self-interest, computational hardness, and the persistence of network effects. Three results follow. First, the payoff advantage of Exit over Stay is strictly increasing in adoption: every payoff term favors Exit, and actors holding meaningful capital see their tipping points fall under structural debasement (Proposition 1). Second, no Stay coalition can hold itself together once any member’s tipping point has been crossed under open access (Proposition 2). Third, above the critical mass p_c no Stay coalition can re-form, and the set of starting points from which adoption reaches the high-Exit configuration grows over time (Theorem 1). Together these establish Bitcoin Exit Dominance (Theorem 2).

1. Introduction

A capital allocator with purchasing power to protect faces a choice between two settlement systems. The first is the prevailing one: sovereign currency, custodied by regulated intermediaries, settled through the banking and securities infrastructure. Call this Stay. The second is a settlement asset that no government issues, no intermediary custodies, and no political body can debase or freeze. Call this Exit. The interesting question is not which a single allocator prefers, but what happens when every allocator faces the same choice at the same time.

The choice became possible on 3 January 2009. Nakamoto²¹'s Bitcoin client mined its first block, and embedded in the coinbase transaction was the front-page headline of *The Times* of London that morning: "Chancellor on brink of second bailout for banks." The block carried no financial information. It carried a pointer to what the proposed system was meant to escape: the bailout regime that absorbs private failures into public balance sheets and pays for them through monetary debasement. The genesis block is the question this paper formalizes. When capturable settlement systems can be debased without recourse, can rational actors settle outside them?

The question is structural, not conjunctural. Global public debt approaches 100% of GDP, with adverse scenarios projecting roughly 115% by the early 2030s (IMF¹⁵); historical repayment of sovereign debt at this scale has occurred through financial repression and debasement rather than taxation (Reinhart and Sbrancia²³). What allocators do against that backdrop, individually and in aggregate, determines whether neutral settlement scales or stays a curiosity.

This paper takes the choice and makes it a game. Each allocator picks Exit or Stay observing the current capital-weighted Exit fraction p_t . Payoffs depend on p_t : as more capital exits, the Exit side becomes more attractive through deeper markets, lower volatility, and normalized regulation; the Stay side becomes worse through rising non-adoption penalties. The question is whether this game has an equilibrium in which p_t stays low forever, oscillates between regimes, or admits a critical mass past which no group of actors has reason to switch back to Stay.

Seventeen years of empirical record point at the third equilibrium. Spot Bitcoin exchange-traded products cleared U.S. regulatory approval in January 2024 (SEC²⁷); BlackRock's iShares Bitcoin Trust accumulated roughly \$37 billion in cost basis within its first fiscal year (BlackRock⁵); central bank gold reserves rose at the fastest pace since the 1970s after the 2022 sanctions on Russian foreign-exchange holdings (Arslanalp, Eichengreen, and Simpson-Bell¹). This paper formalizes the game already in motion.

2. Related Work

The closest prior work is Brunnermeier, James, and Landau⁶, who pose the same coordination problem. Their response — institutional coordination via central bank digital currencies — differs from this paper’s premise (Assumption 1) and fails its test: a CBDC digitizes the capturable system rather than escape it, satisfies neither D1 nor a load-bearing subset of P1–P7, and is an instance of Stay on which Theorem 2 operates.

The existing Bitcoin game-theory literature analyzes a different layer of the same system. Biais, Bisiere, Bouvard, and Casamatta⁴ show that following the longest chain is an equilibrium of the miner game; Eyal and Sirer⁹ characterize selfish mining above a one-third hashrate threshold; Budish⁷ bounds attack cost in network value; and Chen⁸ extends the security-budget argument to fee markets. All four study the miner game. The present paper analyzes the allocator meta-game one layer above: capital holders choose between capturable and neutral settlement. The miner game is proved in prior work and not re-derived here.

3. Model

3.1 Maintained Assumptions

The analysis rests on four assumptions. Each is independently assessable. All subsequent claims derive from these assumptions; rejecting any assumption invalidates the specific claims that depend on it.

Assumption 1 (Multipolarity). No single entity permanently governs global economic activity. Power distributes across competing centers with superlinear coordination costs. The multipolar configuration is dynamic. The institutional-fragmentation index D_t (§3.6) is monotone non-decreasing on the long-run trajectories relevant to Theorem 1.

Assumption 2 (Rational Self-Interest). Actors optimize for self-interest. When defection from cooperative agreements is unpunished or unpunishable, actors defect.

Assumption 3 (Computational Hardness). Certain mathematical problems remain computationally intractable. Digital scarcity and cryptographic custody are implementable.

Assumption 4 (Network Effect Persistence). Past critical mass, switching costs exceed marginal gains of alternatives. Incumbency compounds (Katz and Shapiro¹⁶).

3.2 Definitions

Definition 1 (Neutral Settlement). An asset S is a *neutral settlement asset* if and only if S is immune to capture by force, debasement, or coordinated political action. Table 1 lists the seven properties P1–P7 jointly necessary for D1, alongside the attack class each blocks. Property P6 (neutrality) corresponds to what Huberman, Leshno, and Moallemi¹⁴ term “monopoly without a monopolist”: the protocol exhibits monopoly-like network effects without an entity capable of exercising monopoly power. The cryptographic and consensus primitives that ground P1–P3 and P5–P6 trace to Nakamoto²¹. Bitcoin satisfies P1–P7. The formal definition, irreducibility, and uniqueness of P1–P7 across asset classes are taken up in a companion paper in preparation; the present paper takes P1–P7 as maintained assumptions (§3.2), as it does the miner sub-game result.

Table 1. The seven properties jointly necessary for Definition 1. Each row names what an asset must guarantee and the attack class it forecloses.

Property	Blocks
P1: Informational security	Physical confiscation
P2: Protocol security	Protocol rule change
P3: Permissionless access	Transaction censorship
P4: Absolute scarcity	Supply inflation
P5: Cheap finality	Prohibitive cost/delay
P6: Neutrality	Governance takeover
P7: Adaptive resilience	Obsolescence

Gold is the historical neutral-settlement instrument and remains relevant: central bank gold reserves have risen since the Global Financial Crisis, accelerating in correlation with financial-sanctions risk on the main reserve issuers (Arslanalp, Eichengreen, and Simpson-Bell¹). Density, divisibility, recognizability, and aesthetic appeal have sustained gold as money for five millennia in the market-selection account of monetary emergence (Menger¹⁹). Other proto-monies including wampum, cowrie shells, and salt shared these surface properties but failed P4: industrial production, oceanic shipping, and refined extraction each broke their scarcity. Only gold has held P4 across millennia of technological advance. Gold’s known structural failures are physical seizure, with the 1933 Executive Order 6102 as the canonical precedent; assay risk at unit level, exemplified by tungsten-cored bar fraud; and settlement cost that scales with physical mass. Bitcoin closes these failures while preserving gold’s monetary properties; the formal P1–P7 treatment is taken up in the companion paper in preparation.

Definition 2 (Capturable System). A monetary system in which at least one of P1–P7 fails for its underlying settlement asset.

Definition 3 (Exit). The action of moving capital from a capturable system to a neutral settlement asset; distinct from Hirschman’s¹³ exit/voice/loyalty triad, which describes responses to organizational decline. The complement is *Stay*: holding capital in the prevailing capturable system.

Definition 4 (Critical Mass). The critical-mass threshold at slowly varying state (K, D) is

$$p_c(K, D) := \inf \left\{ p \in [0, 1] : \sum_{i: \Delta_i(p; K, D) > 0} a_i > 0 \right\},$$

where Δ_i , K , and D are defined in §3.6.

3.3 Game

Monetary coordination is modeled here as a non-cooperative game. Each capital allocator chooses one of two actions: hold capital in the prevailing capturable system (*Stay*), or move it to a neutral settlement asset outside that system (*Exit*).

Players. The set N is the population of capital allocators, weighted by their share a_i of total capturable capital. Sovereigns, central banks, corporates, asset managers, and household wealth all participate. Headcount does not move the game; capital does.

Objective. Each player maximizes utility. The Exit side nets real return against volatility, switching cost, and regulatory exposure. The Stay side nets capturable return against capturable volatility and a non-adoption penalty. Players differ in how much they weight each cost, captured by heterogeneity parameters $\lambda_i, \kappa_i, \rho_i$.

Rules. Each player observes the current capital-weighted adoption fraction $p_t \in [0, 1]$ and picks Exit or Stay. As more capital exits, the Exit side becomes more attractive: deeper markets, lower volatility, regulation normalizes. The Stay side weakens through rising non-adoption penalties. Exact payoffs are stated in §3.4; the monotonicity conditions governing how they respond to p are stated in §3.5.

Dynamics. Background conditions evolve slowly. Recognition of the financial-repression channel spreads across capital cohorts, and institutional fragmentation trends upward across multi-decade horizons. Both push individual thresholds down, gradually moving the game toward the critical mass past which Stay coalitions can no longer hold together (§3.6). Section 4 says what makes ignition possible, what blocks ignition, and why Exit, once won, does not reverse.

3.4 Payoffs

The game is $G = (N, S, u)$ in the standard non-cooperative form (Fudenberg and Tirole¹²). The player set $N = \{1, \dots, n\}$ is the population of capital allocators, where actor i holds capital share $a_i \in [0, 1]$ with $\sum_i a_i = 1$. Each strategy set is $S_i = \{\text{Exit}, \text{Stay}\}$. The aggregate adoption process is $p_t = \sum_{i \in \text{Exit}_t} a_i \in [0, 1]$, the capital-weighted fraction at Exit. This dollar-weighted convention follows monetary economics, where reserve composition and capital flows are weighted by stake rather than by headcount: holdouts with negligible capital weight do not move p_t .

Because payoffs depend on others only through the aggregate p , the utility function $u_i : S \rightarrow \mathbb{R}$ is written in two cases:

$$u_i^E(p) := R_B(p) - \lambda_i \sigma_B(p) - \kappa_i K_A(p) - \rho_i R_A(p),$$

$$u_i^S(p) := R_F - \lambda_i \sigma_F - K_N(p).$$

Exit payoff components.

- $R_B(p)$: real return on the neutral asset.
- $\lambda_i \sigma_B(p)$: volatility cost, scaled by personal risk aversion.
- $\kappa_i K_A(p)$: adoption penalty, the switching cost of moving to neutral settlement (Katz and Shapiro¹⁶; Farrell and Saloner¹⁰), scaled by personal career-risk loading.
- $\rho_i R_A(p)$: regulatory penalty, scaled by personal regulatory exposure.

Stay payoff components.

- R_F : capturable return.
- $\lambda_i \sigma_F$: volatility cost on the capturable side.
- $K_N(p)$: non-adoption penalty, a negative network externality on holdouts as competitors exit.

Each actor carries three personal weights: λ_i on volatility, κ_i on the career risk of switching infrastructure, and ρ_i on regulatory exposure. All three are strictly positive and differ across actors, which is why the threshold p_i^* at which Exit becomes attractive varies across the population: a sovereign wealth fund and a retail allocator face the same payoff equation but cross at different p . The functions R_B , σ_B , K_A , R_A , K_N are continuous and bounded on $[0, 1]$. $\sigma_B(p)$ is a quadratic risk measure consistent with mean-variance preferences. Bitcoin's expected return $R_B(p)$ and volatility $\sigma_B(p)$ both depend on adoption because adoption raises the dollar value of the block reward that pays for protocol security (Pagnotta²²; Chen⁸).

Table 2 collects u_i^E and u_i^S in normal form: the row is player i 's action, the column is what the rest of the capital is doing, and each cell is the payoff from the equations above evaluated

at the corresponding p .

Table 2. Normal-form payoff matrix for player i , in the shorthand u_i^E, u_i^S defined above.

	Others: Stay	Others: Exit
Player i : Stay	$u_i^S(0)$	$u_i^S(p)$
Player i : Exit	$u_i^E(0)$	$u_i^E(p)$

3.5 Monotonicity Conditions

Five conditions govern how the payoff primitives respond to rising adoption p .

- (M1) $R'_B(p) > 0$: Network effects increase return (Assumption 4)
- (M2) $\sigma'_B(p) < 0$: Deeper markets reduce volatility
- (M3) $K'_A(p) < 0$: Adoption penalty falls with adoption
- (M4) $R'_A(p) < 0$: Regulatory penalty decreases as adoption normalizes compliance
- (M5) $K'_N(p) > 0$: Non-adoption penalty rises as competitors exit

Each condition specifies a single-direction response to rising adoption. Three concern the Exit side becoming more attractive as adoption deepens: returns rise (M1), volatility falls (M2), and switching costs fall (M3). One concerns regulatory friction normalizing as adoption becomes mainstream (M4). One concerns the externality on non-adopters strengthening as competitors exit (M5). These are empirical conditions, maintained to hold on average over multi-decade horizons. §5 returns to this point.

3.6 Threshold Drift

The capturable system's failure mode is not the overt seizure, capital control, or hyperinflation that history records as discrete events. It is the continuous extraction of real wealth through the interaction of nominal asset appreciation and the taxation of nominal gains. As debasement raises the price level, equities, real estate, and similar capturable assets appreciate nominally to preserve real value. Holders are then taxed on the nominal gain. Post-tax real returns are negative (Feldstein¹¹); the public-debt numerator grows while the real value of private claims is transferred to service it. The decline is imperceptible because nominal balances do not contract. The loss registers only when measured against the price level. This is the channel through which Assumption 1 delivers $R_F < 0$ in real terms, and it is what cohorts come to recognize over multi-decade horizons.

Let $K_t \in [0, 1]$ be the share of capital cohorts that recognize capturable assets as financially repressed rather than risk-free. Recognition need not arrive through a discrete cap-

ture event but accumulates as cohorts observe that nominal returns, after tax on nominal capital gains, fail to keep pace with the price level. Let $D_t \in [0, \infty)$ be an institutional-fragmentation index summarizing four inputs: debt-to-GDP, the inflation–capital-gains tax wedge, financial-repression incidence, and currency-block fragmentation. Both are slowly varying state variables, monotone non-decreasing on the multi-decade horizons relevant to Theorem 1, as observed in post-1970 fiat regime trends (Reinhart and Sbrancia²³).

Each actor’s threshold is a function of the slowly varying state:

$$p_i^*(K, D) := \inf\{p \in [0, 1] : \Delta_i(p; K, D) > 0\}.$$

Assume the sign conditions

$$\frac{\partial p_i^*}{\partial K} \leq 0, \quad \frac{\partial p_i^*}{\partial D} \leq 0.$$

Strict inequality holds on a set of positive capital weight. Higher D_t lowers R_F in real terms. Higher K_t erodes $\kappa_i K_A$ and $\rho_i R_A$. Fewer actors still treat the capturable system as a safe numeraire. Both effects raise $\Delta_i(0; K, D)$ and push p_i^* down.

The critical mass $p_c(K, D)$ defined in §3.2 inherits the same sign structure: rising D_t and rising K_t push p_c down over multi-decade horizons, lowering the bar for ignition. Below p_c no positive-capital-weight set prefers Exit; at or above p_c , defection by some positive-weight set is incentivized. Whether adoption continues or stalls depends on the threshold density of the actor population. If successive thresholds are sparse relative to first-defector capital weights, the cascade may pause. Further drift in (K_t, D_t) resumes ignition. This is the path-dependent adoption mechanism of Arthur²: small differences in early allocation history, compounded by increasing returns to adoption, select the long-run regime.

The model’s claims are structural and directional: existence of thresholds $\{p_i^*\}$, sign of comparative statics, and direction of basin expansion under accumulating (K_t, D_t) . It does not predict specific threshold values, tipping times, or saturation levels, which depend on functional forms and parameter calibrations the model leaves unspecified. This follows the qualitative comparative-statics paradigm of Milgrom and Shannon²⁰ and Topkis²⁶ and the path-dependent lock-in paradigm of Arthur².

An adoption level p is an equilibrium of the Exit Game when no actor, alone or in any group that holds together, gains by switching. §4 establishes that above p_c this holds: the high-Exit outcome survives any returning coalition, and the set of starting points that reach it only grows as (K_t, D_t) accumulate.

4. Results

The argument proceeds in four steps. §4.1 fixes the static incentive: the per-period advantage of Exit over Stay rises with adoption (Proposition 1). §4.2 takes that incentive into the coordination problem: no coalition holding Stay survives unilateral defection (Proposition 2). §4.3 takes the coordination result into the dynamic problem: above the critical mass p_c the high-Exit configuration holds against any defecting group, and under threshold drift more starting points lead to the high-Exit configuration (Theorem 1). §4.4 composes the three into the synthesis claim, Bitcoin Exit Dominance (Theorem 2). Each step uses the previous; none of the three intermediate results is the title claim on its own.

4.1 Payoff Advantage

Hold the slowly varying state (K, D) fixed. The payoff differential $\Delta_i(p)$ rises strictly with p for every player type. Each player has a threshold p_i^* above which Exit is the unique best response. By convention, $p_i^* = +\infty$ for permanent holdouts whose Δ_i stays non-positive on $[0, 1]$.

Proposition 1 (Monotone Exit Advantage). *Under Assumptions 1–4 and maintained conditions (M1)–(M5):*

- (i) *The payoff differential $\Delta_i(p)$ is strictly increasing in p for all player types.*
- (ii) *For each player i there exists a threshold $p_i^* \in [0, 1] \cup \{+\infty\}$ such that Exit is the unique best response for $p > p_i^*$.*

Proof. Define the payoff differential:

$$\begin{aligned}\Delta_i(p) &= u_i(\text{Exit}) - u_i(\text{Stay}) \\ &= [R_B(p) - R_F] - \lambda_i[\sigma_B(p) - \sigma_F] - \kappa_i \cdot K_A(p) - \rho_i \cdot R_A(p) + K_N(p).\end{aligned}$$

Taking the derivative with respect to p :

$$\frac{d\Delta_i}{dp} = R'_B(p) - \lambda_i \cdot \sigma'_B(p) - \kappa_i \cdot K'_A(p) - \rho_i \cdot R'_A(p) + K'_N(p).$$

Under (M1)–(M5), each term is positive (Table 3).

Therefore $d\Delta_i/dp > 0$: the advantage of Exit rises strictly with p . By the monotone comparative-statics results of Milgrom and Shannon²⁰ and Topkis²⁶, each actor's best response is non-decreasing in p .

Table 3. Sign of each term of $d\Delta_i/dp$ under (M1)–(M5).

Term	Sign	Reason
$R'_B(p)$	> 0	(M1)
$-\lambda_i \cdot \sigma'_B(p)$	> 0	$\lambda_i > 0$, $\sigma'_B(p) < 0$ by (M2)
$-\kappa_i \cdot K'_A(p)$	> 0	$\kappa_i \geq 0$, $K'_A(p) < 0$ by (M3)
$-\rho_i \cdot R'_A(p)$	> 0	$\rho_i \geq 0$, $R'_A(p) < 0$ by (M4)
$K'_N(p)$	> 0	(M5)

Define each player's threshold as

$$p_i^* := \inf\{p \in [0, 1] : \Delta_i(p; K, D) > 0\},$$

with the convention $p_i^* = +\infty$ when the set is empty. For actors with $p_i^* \in [0, 1]$, the intermediate value theorem applied to the continuous Δ_i delivers $\Delta_i(p_i^*) = 0$, and for $p > p_i^*$ Exit strictly dominates. Actors with $p_i^* = +\infty$ never defect within the unit interval; the high-Exit region Theorem 1 establishes admits them by construction.

Under Assumption 1, $R_F < 0$ in real terms (see §1; Reinhart and Sbrancia²³). This ensures that even at $p = 0$, the term $[R_B(0) - R_F]$ provides positive contribution, lowering p_i^* toward zero for the actors whose remaining parameters are not extreme.

□

When $R_F < 0$, the §3.4 payoff matrix (Table 2) collapses: for any actor whose threshold p_i^* has been crossed, Exit pays more than Stay regardless of what others do.

4.2 Coordination Failure

Proposition 1 gives the per-actor incentive but says nothing about group behavior. A coalition might still try to hold Stay collectively. It cannot. Standard free-rider logic disqualifies any all-Stay coalition with crossed-threshold members from being self-enforcing.

Proposition 2 (Coalition Instability). *Under Assumption 2 and Property P3, no Stay coalition C containing at least one actor with a crossed threshold ($p > p_j^*$ for some $j \in C$) is self-enforcing. The result is in the spirit of Bernheim-Peleg-Whinston self-enforcement, applied here at the unilateral level: when defection is unpunishable, the crossed-threshold member defects, breaking the coalition at the Nash layer that any CPNE refinement requires.*

Proof. Consider a coalition $C \subseteq N$ maintaining Stay. For this coalition to be self-enforcing in the sense of Bernheim, Peleg, and Whinston³, no member $j \in C$ can profit from unilateral defection.

But by Proposition 1, $u_j(\text{Exit}) > u_j(\text{Stay})$ for j when p exceeds p_j^* . Property P3 (open access) makes the institutional defection cost zero: no counterparty can refuse the trade and no clearing intermediary can foreclose settlement. Sanctions enter the payoff function via R_A and are already priced into each actor's threshold; Proposition 2's claim presumes that pricing.

The coalition faces a standard free-rider problem. The benefit of collective Stay is a public good (maintained value of capturable system); the benefit of individual defection is a private good (captured network effects, avoidance of debasement). Under Assumption 2, the private good dominates.

The dynamics compound: the first mover captures fleeing capital, raising $K_N(p)$ for remaining coalition members and accelerating further defection.

Formally, for at least one $j \in C$,

$$u_j(\text{defect from } C) > u_j(\text{remain in } C),$$

which is sufficient to violate Bernheim-Peleg-Whinston coalition-proofness.

The first-mover advantage is empirically observable: El Salvador adopted Bitcoin as legal tender in 2021 (Republic of El Salvador²⁴), and the U.S. Securities and Exchange Commission approved spot Bitcoin exchange-traded products in January 2024 (SEC²⁷), with BlackRock's iShares Bitcoin Trust accumulating roughly \$37 billion in cost basis within its first fiscal year (BlackRock⁵). $\square$

Remark. Coalitions of permanent-holdout actors ($p_j^* = +\infty$ for all $j \in C$) are admitted by Proposition 1's threshold convention and consistent with the high-Exit region Theorem 1 establishes. Such coalitions, when maintained by coercive sanction against P3, are subject to preference falsification (Kuran¹⁸): publicly observed Stay coexists with privately optimal Exit, and the first-mover advantage operates on private allocations regardless of public posture. P3 reduces the technical cost of private holding to zero. A dimensionless social cost remains for maintaining the public Stay posture while privately exiting. Under the multi-polar enforcement of Assumption 1 (no global authority, jurisdictional arbitrage, imperfect detection), this social cost is bounded for any individual actor. The capital-preservation benefit under $R_F < 0$ is not bounded. Once accumulated private Exits show up in p_t , the public Stay posture no longer holds for crossed-threshold actors.

4.3 Ratchet

This section establishes forward invariance of the absorbing region $\{p > p_c\}$ and monotone basin expansion under §3.6 drift. The argument proceeds by case-split on candidate Stay coalitions: those intersecting the crossed-threshold set $H(p)$ break by Proposition 2; those disjoint from $H(p)$ do not grow.

Theorem 1 (Ratchet Form; cf. Arthur 1989 lock-in²). *Under Assumptions 1–4, maintained conditions (M1)–(M5), and the §3.6 sign conditions: (a) the absorbing region $\{p > p_c\}$ is forward invariant under perturbations remaining within it. (b) Under monotonic accumulation of (K_t, D_t) , the critical mass $p_c(K_t, D_t)$ drifts down, so the basin of local stability expands monotonically over time. The ratchet does not reverse: once p enters the region above p_c , the basin only grows.*

Proof. (a) *Local stability above p_c .* If $p > p_c(K, D)$, the set $H(p) = \{i : \Delta_i(p; K, D) > 0\}$ has positive capital weight. Any perturbation attempting to coordinate a return to high-Stay above p_c is a candidate Stay coalition C and falls into one of two cases. If $C \cap H(p)$ is nonempty, Proposition 2 applies: the crossed-threshold member defects under P3, and C is not self-enforcing. If $C \cap H(p)$ is empty, C consists entirely of uncrossed-threshold capital. These actors are already at their preferred play at the current adoption level and do not move it. Neither case grows. The high-Exit configuration is locally stable. The static part is standard Bernheim-Peleg-Whinston coalition-proofness; the dynamic part, in which the finite-threshold Stay pool shrinks as (K_t, D_t) accumulate, is in the spirit of Konishi and Ray¹⁷. Drawdowns and cascade pauses along p_t are admitted. A permanent stall would require two conditions to hold jointly: p_t plateauing below p_c indefinitely, and p_c failing to fall. The second condition is ruled out by part (b).

(b) *Basin drift.* The §3.6 sign conditions give $\partial p_i^* / \partial K \leq 0$ and $\partial p_i^* / \partial D \leq 0$. The critical mass inherits these signs: $\partial p_c / \partial K \leq 0$ and $\partial p_c / \partial D \leq 0$. Under monotonic accumulation of (K_t, D_t) , $p_c(K_t, D_t)$ is non-increasing in t . The basin of local stability $\{p : p > p_c(K_t, D_t)\}$ therefore expands monotonically. The ratchet structure is the lock-in by historical events of Arthur² applied to monetary adoption: not pointwise irreversibility, but monotone expansion of the locally stable absorbing region as (K_t, D_t) accumulate. $\square$

The high-Exit region above p_c allows partial adoption, in the manner of incumbent network goods such as USD reserve status, English as lingua franca, or Google search dominance, which empirically plateau at finite shares rather than universal adoption (Katz and Shapiro¹⁶; Farrell and Saloner¹⁰). Permanent holdouts are admitted by construction. Some Stay capital remains, parked at a focal coordination level (Schelling²⁵) that no returning group can unwind. Adoption settles short of universal Exit.

4.4 Bitcoin Exit Dominance

Proposition 1 gives the static incentive. Proposition 2 rules out Stay coalitions with crossed-threshold members. Theorem 1 gives local stability above p_c with monotone basin expansion. The realized process exhibits all three at once. That conjunction is the title claim.

Theorem 2 (Bitcoin Exit Dominance). *Under the joint hypotheses of Propositions 1–2 and Theorem 1, the realized capital-weighted adoption process satisfies all three of:*

- (i) *the payoff advantage of Exit over Stay is strictly increasing in p for every player type (Proposition 1);*
- (ii) *no Stay coalition with at least one crossed-threshold member is self-enforcing (Proposition 2);*
- (iii) *the high-Exit configuration is locally stable above p_c , and the basin of local stability expands monotonically under threshold drift (Theorem 1).*

Proof. Proposition 1 gives (i) directly under (M1)–(M5). Proposition 2 gives (ii) under Assumption 2 and Property P3, with (i) supplying the per-defector incentive that breaks any Stay coalition with at least one crossed-threshold member. Theorem 1 gives (iii): local stability of the high-Exit configuration above p_c via Proposition 2 applied to candidate Stay coalitions, and basin expansion via the §3.6 sign conditions on $p_c(K_t, D_t)$. The conjunction of (i), (ii), and (iii) is the Bitcoin Exit Dominance claim: strictly increasing payoff advantage in p , instability of crossed-threshold Stay coalitions, and a locally stable high-Exit absorbing region whose basin expands under threshold drift. $\square$

Remark. By the implicit function theorem applied to $\Delta_i(p_i^*) = 0$,

$$\frac{dp_i^*}{d\lambda_i} = -\frac{\partial\Delta_i/\partial\lambda_i}{d\Delta_i/dp} = \frac{\sigma_B(p_i^*) - \sigma_F}{d\Delta_i/dp},$$

which is positive when $\sigma_B(p_i^*) > \sigma_F$, the empirically relevant case of a higher-volatility neutral asset relative to the capturable side. Under that condition, more risk-averse actors carry higher thresholds and cross them later; because $d\Delta_i/dp > 0$, they cross once p rises sufficiently.

5. Limitations

The paper takes P1–P7 as maintained assumptions on Definition 1. Their formal characterization, irreducibility, and uniqueness across asset classes are the subject of a companion

paper in preparation; the present paper does not establish that no alternative asset satisfies P1–P7 jointly.

The model’s claims are qualitative and directional: thresholds $\{p_i^*\}$ exist and fall as (K_t, D_t) accumulate, and the set of starting points reaching the high-Exit region grows over time. It does not predict specific threshold values, tipping times, or saturation levels.

The monotonicity conditions (M1)–(M5) are stated to hold on average over multi-decade horizons rather than at every instant. Transient violations, for instance $R'_B(p) < 0$ during a network-congestion crisis, are within scope.

Adoption-timing phenomena are outside scope: price declines, regulatory actions against specific intermediaries, developer controversies, energy debates, and individual-cohort deleveraging affect when, not whether, the high-Exit region is reached.

6. Conclusion

The standard frame for monetary coordination treats sovereign issuers as the only agents that matter. The Exit Game inverts the frame: capital allocators decide what counts as settlement, and the result is a game whose long-run Exit outcome depends on the existence of an asset that no government issues. Once such an asset exists and satisfies P1–P7, the question is no longer whether allocators can leave the capturable system but whether continued debasement and fragmentation push adoption past p_c .

The implication for the international monetary system is that reserve composition depends on settlement-system characteristics rather than only on political coordination. Central bank gold accumulation since the Global Financial Crisis is the directional precedent. The Exit Game predicts that any settlement asset which closes gold’s three structural failures of assay, confiscation, and supply growth pulls capital the same direction at lower cost; the §3.6 threshold-drift mechanism predicts that continued financial repression and institutional fragmentation keep pushing more actors into Exit.

Two questions are deliberately deferred. Whether neutral settlement is unique to Bitcoin, or some other asset class can also satisfy P1–P7 jointly, is the subject of the companion paper in preparation. Whether the allocator population can be extended to include autonomous economic agents whose strategic structure differs from that of human capital allocators is the subject of the second.

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Notation

Table A1. Notation used throughout.

Symbol	Definition
G	Settlement game (N, S, u)
N	Set of capital allocators
S_i	Strategy set: {Exit, Stay}
u_i	Utility function
a_i	Actor i ’s share of total capturable capital, $\sum_i a_i = 1$
p_t	Aggregate capital share at Exit at time t : $\sum_{i \in \text{Exit}_t} a_i$
$p_i^*(K, D)$	Threshold for player i at slowly varying state (K, D)
$p_c(K, D)$	Critical-mass threshold at (K, D)
K_t	Financial-repression recognition share, $K_t \in [0, 1]$
D_t	Institutional-fragmentation index, $D_t \in [0, \infty)$
$\Delta_i(p; K, D)$	Payoff differential: $u_i(\text{Exit}) - u_i(\text{Stay})$ at (K, D)
$R_B(p)$	Expected real return on neutral settlement asset
R_F	Expected real return on capturable assets
$\sigma_B(p)$	Volatility loading on neutral settlement asset
σ_F	Volatility loading on capturable assets
$K_A(p)$	Adoption penalty
$R_A(p)$	Regulatory penalty (decreases as adoption normalizes compliance)
$K_N(p)$	Non-adoption penalty
$\lambda_i, \kappa_i, \rho_i$	Risk aversion, career penalty, regulatory penalty (all > 0 strictly)
(M1)–(M5)	Monotonicity conditions (§3.5)

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